



COMSATS University Islamabad

Department of Computer Science

Course Description Form (CDF)

Course Information

Course Code: **CSC270**

Credit Hours: **4(3,1)**

Lab Hours/Week: **3**

Course Title: **Database Systems**

Lecture Hours/Week: **3**

Pre-Requisites: **None**

Course Objective

- To explain the basic database concepts, information retrieval and relational theory;
- To develop the relational data model;
- To develop an enterprise data model that reflects the organization's fundamental business rules;
- To apply normalization techniques;
- To discuss the basics of transaction management, concurrency controls, query mechanisms, security and quality issues;
- To apply database programming languages and physical database design to gain experience in term project.

Course Content

This course introduces the fundamental concepts of database systems. Topics include: Introduction to Databases & Information Systems; Evolution of Database Systems; Components; Architecture; Functions; Relational Model; Relational Algebra; Relational Calculus; Data Modeling; Relational Data Model; Relational Algebra & Calculus; Integrity Constraints; Conceptual Models; Entity-Relationship (E-R) Model; Enhanced E-R Model; Mapping Conceptual Schema to Relational Schema; Functional Dependency & Normalization; Structured Query Language (SQL); Views; Materialized Views; Non-Relational/No SQL Databases; MongoDB as NoSQL Database; Document Model; and Transaction Management.

Unit wise Major Topics:

Unit	Topic	No. of Teaching Hours
1.	Databases & Information Systems: Basic Concepts, Evolution, Advantages & Disadvantages, Database Environment, Approaches & Roles; Components; Architecture; Functions; Data Dependences, and Applications.	6
2.	Relational Model: Relation, Domains, Properties, Attributes; Relational Keys; DB Schema; and Integrity Constraints.	3
3.	Relational Algebra (Operations): Unary, Set, Join, Division, Aggregation & Grouping; and Relational Calculus: Domain & Tuple.	6
4.	Data Modeling: Conceptual Models, Relational Database Design; E-R Model; and Enhanced E-R Model.	9
5.	Normalization: Purpose, Data Redundancy & Anomalies; Process: First Normal Form, Second Normal Form, Third Normal Form, and Boyce-Codd Normal Form.	7.5
6.	SQL: DDL, DML, DCL; Views; Materialized Views; and Database Authorization.	4.5
7.	Non-relational / No SQL Databases: MongoDB as NoSQL Database; MongoDB Sharing & Replication; and Document Model.	4.5
8.	Transaction Management: Support, Failure & Recovery, Concurrency	4.5

	Control, and Transaction & Storage.					
Total Contact Hours				45		
Mapping of CLOs and SOs						
Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	GA		
CLO's for Theory						
CLO-1	1	Explain database concepts and principles.	Understanding	2		
CLO-2	2-3	Apply the concept of domain and tuple relational calculus.	Applying	2,3		
CLO-3	4-5	Apply data modeling and normalization techniques to design database for small to medium size enterprise	Applying	3,4		
CLO-4	8	Describe the principles of transaction management.	Understanding	2		
CLO's for Lab						
CLO-5	6-7	Apply data processing operations on both relational and non-relational DBMS.	Applying	3,5,10		
CLO-6	1-8	Develop a database system for medium size enterprise in a team environment.	Creating	1-6, 10		
CLO Assessment Mechanism						
Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5	CLO-6
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4	-	-
Assignments	-	Assignment 1	Assignment 2-4	-	Lab Assignments	-
Mid Term Exam	Mid Term Exam	Mid Term Exam	Mid Term Exam	-	Lab Mid Term Exam	-
Final Term Exam	Final Term Exam				-	Lab Project/ Lab Final Term Exam
Text and Reference Books						
Textbooks:						
1. Database systems: A Practical Approach to Design, Implementation, And Management, Thomas Connolly y, Carolyn Begg, Pearson, 2015.						
2. MongoDB: The Definitive Guide, Shannon Bradshaw, Eoin Brazil, Kristina Chodorow, O'Reilly Media, 2019.						
Reference Books:						
1. Fundamentals of Database Systems, Elmasri, R, Navathe, Pearson, 2016.						
2. Database System Concepts, Silberschatz, Korth, Sudarshan, McGraw Hill, 2019.						